Technical Data Sheet

Lignostab® 1198 L

lignin stabilizer

general

Lignostab® 1198 L is a photo stabilizer solution developed specifically for wood pre-treatments and effectively stops the photo-oxidation of lignin. It is highly effective for the color stabilization of natural and tinted or stained wood for interior application. Used in wood primers for exterior applications it can extend the retention of properties such as coating adhesion and resistance to failures such as flaking and cracking, therefore resulting in higher durability. Especially for high-performance outdoor applications where long-lasting coatings are required, e.g., wooden window frames, Lignostab® 1198 L also allows the use of non- or low-pigmented coatings.

key benefits

- ready-to-use lignin stabilizer solution for direct wood impregnation
- improved color retention of natural pale and tinted wood (e.g., pine, spruce, maple, ash, birch, …) for interior application
- improves properties such as coating adhesion and resistance to failures such as flaking and cracking for exterior applications (e.g., joinery)

chemical nature

20 % solution of a tetramethyl piperidine derivative

Properties

physical form

orange to red liquid

storage

When kept in original unopened containers and at temperatures of 5 – 35 °C (41 – 95 °F), Lignostab® 1198 L can be stored for up to 6 months from the date of manufacture.

typical properties (no supply specification)

density at (DIN 51757)	20 °C	(68 °F)	0.98 – 1.02 g/cm³

miscibility

easily miscible with water and polar solvents

Application

fields of application

Applications which can benefit from the Lignostab® 1198 L concept are in general all areas where light to pale wood species (or wood-composite materials) with clear or semi-transparent coatings are used. This includes indoor parquetry and furniture as well as outdoor wood and timber work (doors, windows, cladding, paneling, fencing, decking, shingles or beams).

- interior clear wood primers and sealers
- industrial and DIY wood stains and tinting solutions
- exterior wood stains and wood preservative treatments
- sap stain protectants, wood preservers and waterproofers
- penetrating clear and pigmented finishes (i.e., deck coatings) wood care products, waxes and polishes
- wood veneers for automotive interior applications

According to the lignin stabilizer concept an optimized photo-stability can be achieved when Lignostab® 1198 L is applied directly onto the wood substrate by simple aqueous, hydro-alcoholic or hydrocarbon pre-treatment or used in an already utilized impregnation, i.e., in a blue-stain inhibiting primer. Lignostab® 1198 L always has to be combined with selected UV absorbers (for interior applications) or UV absorber/HALS (for exterior) in the same treatment (internal UV filter effect) or, better, in a subsequently applied coating (external filter effect). For aqueous systems (primers and/or top coats), Tinuvin® 99-DW or Tinuvin® 477-DW are the UV absorbers of choice. Tinuvin® 477, Tinuvin® 99-2 or Tinuvin® 5060 may be used in solvent-based alkyd and PUR-based coatings. For more details related to the use of UV absorbers for wood coatings please refer to the technical data sheets.

For tinted wood, Lignostab® 1198 L can be used directly within the colorant-containing tinting solutions or stain providing improved lignin as well as colorant photo-stabilizing properties.

Lignostab® 1198 L does not interact with and does not influence the performance of wood biocides (e.g., iodopropynyl butylcarbamate [IPBC], tebuconazole, propiconazole, dichlofluanide, toluylfluanide), binder materials, rheology modifiers or any other paint ingredients. However, the stability of Lignostab® 1198 L in stains and impregnations should always be checked by careful preliminary testing when long-term storage stability is required.

Cure retardation may occur when UV-curable systems and oxidatively drying alkyd coatings are applied directly to wood treated with Lignostab® 1198 L. The use of a sealer to avoid its migration and/or an adjustment of the photo initiator or siccative composition might overcome this problem.

recommended concentrations

The concentration of Lignostab® 1198 L depends on the pigmentation of the coating above, the desired degree of protection and the wooden substrate. The amount of light stabilizers required for optimum performance should be determined in trials covering a concentration range. Recommended concentrations of Lignostab® 1198 L in pre-treatments in relation to the solvent used:

pre-treatment solvent	Lignostab® 1198 L concentration (by weight on total formulation)
aqueous/alcoholic interior	4.8 – 7.5 %
aqueous/alcoholic exterior	5.0 – 10.0 %
non-polar interior/exterior	10.0 – 15.0 %

The Lignostab® 1198 L pre-treatment can be applied by brush (1 x 80 – 120 g/m^2), curtain coating or dip coating (contact time 10 – 20 s).

guideline formulations

pale wood color protector for interior applications

	% by weight
acrylic binder	30.0 – 40.0
Lignostab® 1198 L	4.0 – 7.5
Efka® 3570 (substrate wetting)	0.2 – 0.5
Efka® 2527 (defoamer)	0.2 – 0.4
Tinuvin® 477-DW	4.0 – 6.0
AMP-90™[1] (adjust to pH 8 – 9)	to be adjusted
water, deionized	to be adjusted
	100.0

aqueous clear impregnation primer for exterior applications

acrylic binder	20.0 – 30.0
Lignostab® 1198 L	5.0 – 10.0
Efka® 3570 (substrate wetting)	0.2 – 0.5
Efka® 2527 (defoamer)	0.2 – 0.4
IPBC (fungicide)	0.2 – 0.4
tebuconazole (fungicide)	0.4 – 0.6
propiconazole (fungicide)	0.4 – 0.6
permethrin (insecticide)	0.1 – 0.2
AMP-90™[1] (adjust to pH 8 – 9)	to be adjusted
water, deionized	to be adjusted
	100.0

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Safety
When handling this product please comply with the advice and information given in the safety data sheet and observe protective and workplace hygiene measures adequate for handling chemicals.

Note
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